

mHCI

Lecture 8: Online Privacy Worldviews on “valid” research

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Leakiness and Creepiness in App Space: Perceptions of Privacy and Mobile App Use

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CHI 2014

The Privacy Paradox

Contradiction between users' *reported* privacy concerns and their observed *behaviour* regarding privacy practices

i.e., apps feel *creepy*, but people use them anyway

Example: they worry about who may get access to their data, but they don't read privacy policies when installing apps

Personal space and territoriality

Altman's privacy regulation theory: privacy as a process through which "people selectively adjust access to themselves through controlling the information they disclose or receive. In other words, **people keep secrets** and manage themselves and their secrets through ongoing **acts of revealing and concealment**".

Personal space: "boundary around a person, intrusion into which is often uncomfortable and generally not permitted". Intrusions or invasions of personal space represent privacy violations.

Territorial behavior: Entry into another person's primary territory is typically done with permission only and trespassing is considered an invasion that can elicit intense emotional responses and physical action.

Smartphones as extensions of “the self”

Smartphones expand users’ “personal space”: their functions are not just about practical problem solving, but also about projecting and constructing the self. They can feel like intimate zones and extensions of users' bodies.

Intrusion into users’ smartphones feel as violations to their personal space, i.e., as **privacy violations**

Example: people regard their phone as highly private and are reluctant to share it with others, expressing discomfort even when sharing it with close friends.

Research Questions

How smartphone users feel about data access on their phones?

Are smartphone users willing to change their behavior once they have been informed about tracking and *data leakage*?

Research Methods

Longitudinal Interview Study (13 participants) + Survey (272 participants)

Study 1 - Interviews (and observations)

Participants:

6 from Iceland and 7 from Denmark. 8 men, 5 women

Recruitment: own personal networks, advertising on Facebook

Inclusion criteria: owned and used Android smartphones, between 25 and 60 years old

Procedure: The interviews were structured around tasks

Task 1 (Think aloud protocol): downloading an application of their choosing.

Task 2: download and try out the game "Fruit Ninja", then discussed its main functions and required permissions

Follow up interview 3 weeks later: Did participants change their behaviour towards protecting themselves from “data leaks”?

Data collected: whether they had uninstalled apps in the last 3 weeks.

Study 2 - Survey

Participants:

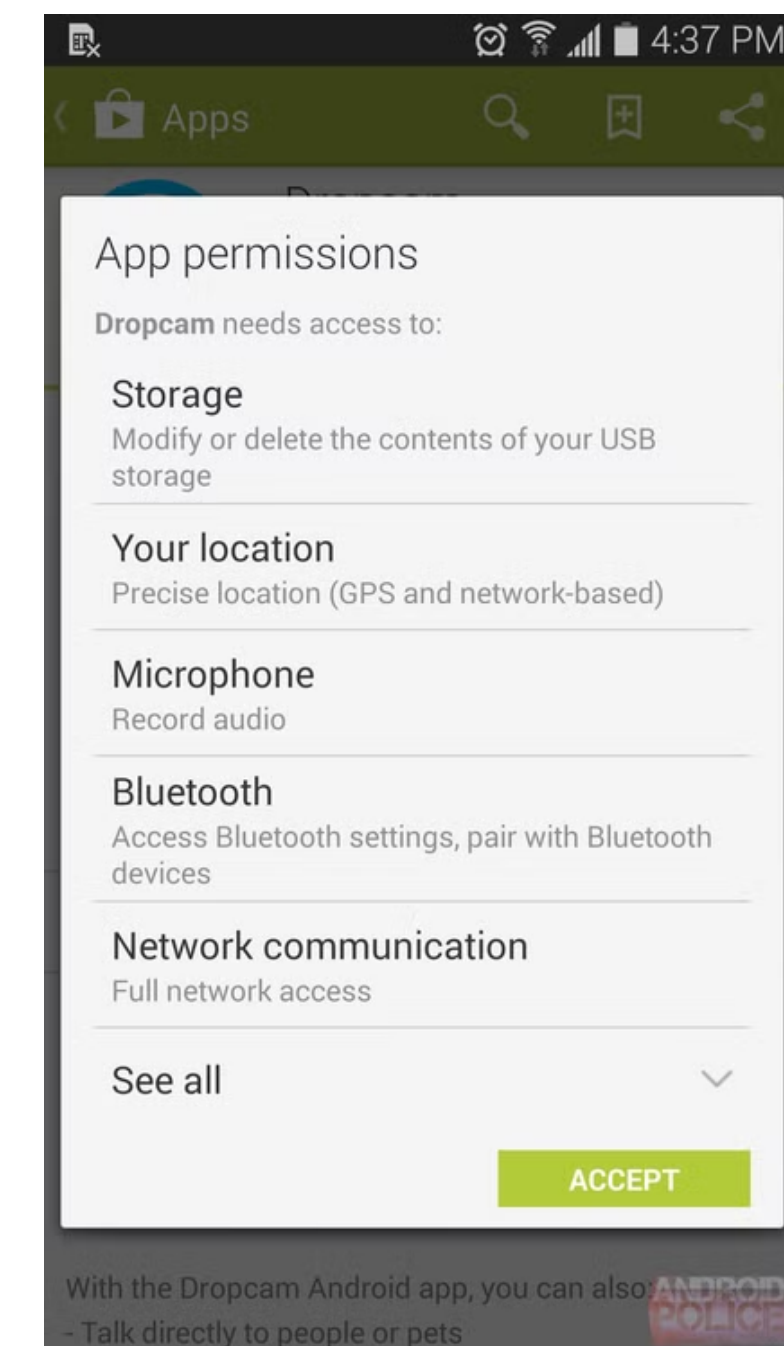
272 participants from Iceland and Denmark.

Recruitment: own personal networks, advertising on Facebook, student mailing lists

Procedure:

First rate appropriateness of the permissions requested by the app (Fruit Ninja or a Flashlight app).

Then, show the receivers of the data collected according to the privacy policies, and ask questions about awareness and feelings.



Example of what the permissions information section looked like on Google Play in 2014, <https://www.androidpolice.com/2014/06/10/simplified-permissions-ui-in-the-play-store-could-allow-malicious-developers-to-silently-add-permissions/>

Why Fruit Ninja and the Flashlight app?

- They collect data that goes beyond what one may find reasonable for the purpose they serve
- The collected data is sold to an ad and analytics company that builds user profiles on aggregated data

Imagine a smartphone owner who has on her phone a typical flashlight application like the High-Powered Flashlight from iHandy inc., and a simple one-player game application like Fruit Ninja from Halfbrick Studios. Both apps require access to the phone number and device ID, the remote number if a call is active and which other applications are used on the device. Additionally the flashlight app has access to precise location, hardware controls and system tools that enable it to change system configuration and display settings. The Fruit Ninja app also requires access to the list of accounts known by the device. The information these apps collect is sold to several third parties, one of which is Flurry, a big ad and analytics company that according to its website prides itself on taking in two terabytes of data from 2.8 billion app sessions per day [11]. Flurry buys data from both these apps and can build a profile of a user that includes information about what type of phone she has, what applications she uses and how frequently, what accounts she has on her phone, which numbers she calls (which can link to other consumer profiles in the Flurry database) and precise location of the phone which, if logged with timestamps, gives an accurate estimate of where she lives, works and travels. By properly aggregating this information the company gets a detailed profile of her as a consumer, information that is worth money.

Findings

More than half (7 of 13) had deleted an application **although in most cases it was Fruit Ninja** (which had not been their own choice):

“I will never waste my time reading privacy policies, my time is simply too valuable. To some extent I just have to accept this.” – interview, Iceland

Just like the participants in [10] and [15] our interviewees ignored the EULA when they were asked to install an application of their own choosing from the app store during the interview. Yet they had typically expressed concern about their privacy in the prior portion of the interview, discussing privacy implications of smartphone application data leakage. When asked why they had ignored the EULA two explanations dominated similar to those described in [9,18,25]. First, they had never had any real negative consequences from data collection. Second, the desire to have the application trumped any concerns for data collection, which was explained away as “the way things are.”

“Yes, I used to think about it, but I decided to stop it. I know I am agreeing to this (data collection). If I really want the app, I take my chances.” – interview, Denmark

all permissions requested by the application. Similar to [25], this exercise revealed a mismatch between the interviewees’ mental models of what Fruit Ninja might reasonably request in order to function and the kind of data it actually requested. That is, they expected the data collected via the app to be contextually appropriate to a single-player game of slicing fruit. Not surprisingly, this mismatch caused participants significant discomfort similar to [9,24]:

“I feel deceived, I had no idea it was this systematic” – interview, Iceland

“I deleted Fruit Ninja because it was gathering a lot of information, although it might also be the case with other apps on the phone I don't want to delete them, I simply enjoy them too much.” – interview, Iceland

Findings

Theme	Code	N
Tracking is understandable	Collecting data is OK	1
	Developers need it	6
The concerned and the outraged	Should be illegal	8
	Users need to be informed / more control	7
Dejected acceptance and compliance	Nothing is free	6
	The user has to comply	14
Tracking is Disturbing/Creepy		50
Unconcerned "I do not care"		2

Table 1: Thematic coding of qualitative data from the survey (total N=89, some statements were coded in more than one category)

“I believe the adage: "When you think you're getting a free lunch; you're actually being served to someone else." It's a trade-off for free entertainment.” – survey, USA

“This is just how these things develop and this will only get worse, we can not change it. But I don't see how I can be personally affected by them knowing stuff about me.” – interview, Denmark

The level of resignation was striking in most of these responses. More often than not, the explanation for why our participants didn't spend the time to read the EULA, overlooked that some of their favorite applications leak data and ignored the assortment of tools available for limiting data leakage of this kind was one of helpless compliance:

“It's unsettling and not ok, but I feel very powerless against it.” – survey, Denmark.

“I don't know what can really be done [about data collection] because then you have to stop using the smartphone as a smartphone.” – interview, Denmark

Discussion and Conclusions

New explanations to the privacy paradox:

Contradictory decisions about online privacy go beyond situational constraints or lack of motivation. They point to **learned helplessness**.

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Repeated invasions into a persons' privacy and a conviction that there is no recourse can result in **learned helplessness, when people stop responding to invasions even when presented with ways to defend themselves.**

significant factor in mental health. Learned helplessness typically happens when people come to believe that a situation is unchangeable or inescapable and will often construct reasons for why this is so even if solutions become available later on. Consider the following statements:

“I silently accept it. When you make me think about it, I kind of don't like it, but have probably forgotten all about it next time I download an app.” – survey, Denmark

“It seems like a necessary evil at this point. Because it is so ubiquitous, I think that it's likely that this sort of thing will never go away.” – survey, USA

In both cases there is an implicit agreement that the respondent has no way of affecting the situation and must accept it if they are to be able to go on. The implication here is that perhaps there are other explanations for what has been termed the privacy paradox [36], beyond decision-making conundrums and situational constraints.

Personal opinions and takeaways

- The study shows an interesting example of *probing* for both attitudinal data (e.g., privacy expectations and concerns) and behavioural data (e.g., uninstalling apps)
- Ethical concern: asking participants to download an app that researchers KNOW that “leak data” to third parties
- **Learned helplessness** could be (is?) misused as a **dark pattern** to sustain invasive data collection practices despite users privacy concerns
- Great example of exploratory research that leads to a data-grounded **conceptual** contribution

What makes a study “valid”?

What makes a study “valid”?

It depends on what you consider “valid”, actually!

8. LIMITATIONS

Qualitative studies with small sample sizes have limitations, such as lack of statistical power. However, the in-depth interviews did allow us to get qualitative insights into participants’ reactions to the notifications and visualizations, as well as their understanding of the information they were seeing. While we provided the numbers of participants that made similar statements, we do not claim that these could generalize to larger populations.

A lab study has limited ecological validity. The participants were not using their own phones, playing games of their own choice, and were not in privacy-sensitive locations (e.g., at home). With a greater range of locations, applications, and situations, participants may be more sensitive to the particular context of information sharing. In real settings, users may actually be more concerned with privacy leakage than when they are in a lab and using a lab phone. We did ask participants to imagine their friend or family member playing the game “on the bus, at the doctor’s office, or waiting to meet you somewhere.” However, we have no indication that participants considered the privacy-sensitive nature of any of these locations.

On the other hand, lab participants may exaggerate their concern and interest in the task at hand (understanding or caring about privacy leakages) to appear to be a good participant. Also, naming our app as Privacy Leaks may have biased some participants who would not have considered privacy risks otherwise. Furthermore, they may not actually do what they claim they will do when they are in the lab. Previous work has shown that survey participants may report

What should the research goal be to consider “lack of statistical power” a limitation to validity?

Why don’t they want to “claim” this?

What do they mean by “ecological validity”?

What do they mean by bias?

Social acceptability / participant bias

Balebako et al., Little Brothers Watching You

There are different “worldviews” regarding what constitutes rigorous, valid research

The worldview that is concerned with producing generalizable, objective findings is called “positivism”

- Goal: to explain observable phenomena (e.g., cause-effect relationships), to predict phenomena and behavior, to discover “universal laws”
- What is good research here?
 - Results that are **generalizable**, for example, to the population represented by the study participants, to other contexts, real life, etc.
 - Research that aims to control variability to produce objective results, compensating for potential biases and limitations to validity. “Validity” is often seen as:
 - **Internal validity:** limitations to the trustworthiness of the results respect to the study’s research goals/questions. For example: an experiment design that doesn’t account for order effects and learning effects may be seen as internally invalid
 - **External Validity:** limitations to the generalizability of the results to other contexts than the one represented in the study (other populations, tasks, devices, cultures, etc). For example: if the study participants are all male, the results shouldn’t claim that they apply to all people.
 - **Ecological Validity** (a special case of external validity): limitations to the generalizability of the results to real-life contexts.

Limitations to Validity within a positivist worldview

Limitations to internal validity

i.e., limitations to the trustworthiness of the results
respect to the study's research goals/questions

Limitations to external validity

i.e., limitations to the generalizability of the results to other
contexts (populations, tasks, devices, cultures, etc)

Limitations to ecological validity

i.e., limitations to the generalizability of the results
to real-life contexts

Limitations

Conducting a field study enabled us to evaluate our privacy nudges in-situ on participants' own devices. This increased ecological validity but introduced multiple challenges. First, we were unable to recruit a larger number of participants, because carriers (e.g., AT&T) and OEMs (e.g., Samsung) rolled out updates to Android 4.4.2 (i.e. AppOps removed) around the same time, which significantly shrunk the pool of potential participants. Second, our study client required Internet connectivity. However,

some participants deactivated data connection to conserve data volume or had intermittent connectivity for other reasons. This affected our data collection and caused some nudges to be lost or delivered later than scheduled. We implemented a monitoring tool to identify participants whose devices were not sending back information regularly, and then reminded them via email to remain connected. While this worked, we used this approach sparsely to avoid biasing participants' responses. Due to these technical difficulties, some participants received all eight nudges while others received fewer nudges. In addition, we acknowledge a potential self-selection bias in our participant pool. These limitations may have skewed our results to be more conservative, but we are confident that they did not undermine our findings in general.

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There are different “worldviews” regarding what constitutes rigorous, valid research

Another worldview is “*interpretativism*”

- Goal:
 - to gain in-depth understanding of how individuals and groups create, interpret, and experience their subjective realities
 - to understand the meaning of human behavior and experiences
 - e.g., propose new concepts (as in the Leakiness and Creepiness paper) that help describe why people act the way they do, rather than focusing on predicting their behavior
- What is good research here?
 - **Reflexivity:** The researchers **acknowledge** their subjectivities! Results aren’t expected to be “objective”; instead, the researchers that interpret the data explain how their backgrounds, experiences and values may shape their interpretation.
 - **Transferability:** Other researchers/designers can “transfer” insights to other contexts. Instead of claiming that the insights themselves generalize across contexts, transferability leaves it to the reader of the insights to decide if they’re helpful to understand similar situations/context
 - **Credibility:** The interpretations are coherent and clearly grounded in the data (i.e., they don’t look “disconnected” from the data)

Be aware of the “worldview” behind each paper and your own!

Otherwise, you risk considering some insights “invalid” while they are actually coming from a different perspective where “internal/external validity” doesn’t make sense